

Anomaly Detection and Representation Learning in U.S. Retail Electricity Markets: A Comparative Study of AutoEncoders, Variational AutoEncoders, and Isolation Forest

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Abstract—This work extends the unsupervised analysis of U.S. retail electricity sales data from Challenge 5 into the domain of deep reconstruction-based anomaly detection and representation learning. Using a 12-feature matrix of monthly state-sector observations, we train AutoEncoders (AE), Variational AutoEncoders (VAE), and an Isolation Forest (IF) baseline to identify anomalous electricity-consumption patterns.

An architecture ablation over latent dimensions and a β sweep for the VAE are performed across multiple random seeds. Detector agreement is evaluated through Spearman rank correlation, while latent representations are visualised with t-SNE and UMAP using the cluster labels obtained in Challenge 5.

Results suggest that reconstruction-based methods identify a subset of anomalous records associated with winter demand peaks and recent high-price consumption regimes that were not fully isolated by partition-based clustering methods. In addition, the VAE latent space improves cluster separability relative to the original feature representation, indicating that deep representation learning captures meaningful geometric structure in the electricity-consumption manifold.

Index Terms—Anomaly Detection, AutoEncoder, Variational AutoEncoder, Isolation Forest, Representation Learning, t-SNE, UMAP, Electricity Consumption, EIA, Unsupervised Learning

I. INTRODUCTION

Detecting anomalous observations in large administrative datasets is a fundamentally different task from discovering the dominant cluster structure. Whereas clustering methods partition the *full* data manifold into coherent groups, reconstruction-based anomaly detectors specifically characterise the *tail* of the distribution: observations that cannot be faithfully compressed and reconstructed by a model trained on the majority of the data. This distinction is operationally important in the energy domain, where a consumption spike caused by an extreme weather event or an industrial shutdown occupies the *interior* of the majority cluster in feature space yet is a qualitatively exceptional event from a grid management and policy perspective.

The U.S. Energy Information Administration (EIA) publishes monthly retail electricity sales disaggregated by state and customer sector, producing a rich longitudinal record that spans 62 states and territories from January 2001 to January 2026. Challenge 2 demonstrated that 10% labeled data is sufficient for Random Forest to classify high-consumption

months with 92% accuracy, confirming that the consumption signal is strong and structured. Challenge 5 established that the dominant unsupervised structure is *scale-based*: a bimodal partition separates the handful of mega-states (Texas, California) from the rest, with DBSCAN additionally flagging price anomalies in Hawaii and Alaska as noise points. Neither analysis, however, characterised the tail behaviour of the reconstruction error distribution or the sub-structure within the dominant clusters, motivating the present deep-learning approach.

AutoEncoders [1] learn a compressed bottleneck representation by minimising reconstruction error; data points that are poorly reconstructed are considered anomalous because they deviate from the learned data manifold. Variational AutoEncoders [2] extend this by imposing a Gaussian prior on the latent space via the Evidence Lower Bound (ELBO), yielding a smooth, continuous manifold suitable for interpolation and visualisation. The β -VAE variant [3] introduces a tunable weight β to trade reconstruction fidelity for disentanglement. Isolation Forest [6] provides a computationally efficient, tree-based classical baseline.

The main contributions of this paper are:

- (i) A fully reproducible AE and VAE pipeline for tabular energy data with multi-seed evaluation and architecture ablation over latent dimensions and β values;
- (ii) A comparative anomaly-detection study across three unsupervised detectors with Spearman rank correlation analysis of detector agreement;
- (iii) Latent-space visualisations (t-SNE and UMAP) connecting deep representations to the Challenge 5 K-Means cluster labels; and
- (iv) A cross-challenge synthesis spanning Challenges 2, 5, and 6 that articulates what each unsupervised paradigm reveals about the structure of U.S. electricity consumption.

II. RELATED WORK

Deep anomaly detection using reconstruction-based models has been surveyed comprehensively by Pang et al. [8], who establish that AutoEncoders outperform classical methods on structured tabular data when the majority class is sufficiently

large to train a representative model. The foundational AE architecture dates to Hinton and Salakhutdinov [1], who demonstrated that deep encoders can learn compact codes qualitatively superior to PCA for high-dimensional data.

The VAE of Kingma and Welling [2] recast AE training as approximate Bayesian inference, regularising the latent space via a KL-divergence penalty against a standard Gaussian prior. An and Cho [7] showed that the VAE reconstruction probability furnishes a principled anomaly score that outperforms the standard AE reconstruction error in several benchmarks. The β -VAE [3] demonstrated that increasing β above unity encourages disentangled latent representations at the cost of reconstruction fidelity, a trade-off directly relevant to our β sweep.

For dimensionality reduction and visualisation, t-SNE [4] and UMAP [5] have become the standard tools for inspecting high-dimensional latent spaces; UMAP additionally preserves global topological structure better than t-SNE at comparable computational cost.

Isolation Forest [6] isolates anomalies by exploiting the observation that rare observations require fewer random binary partitions to be separated from the rest of the data. It remains a strong baseline on tabular datasets, as confirmed by the survey of Goldstein and Uchida [9].

In the energy domain, autoencoders have been applied to smart-meter time series for non-technical loss detection and outage identification [10]. The present work differs in that it operates on aggregated monthly state-sector statistics, providing a meso-scale view of U.S. consumption structure rather than individual customer behaviour.

III. DATA AND PREPROCESSING

A. Dataset

The dataset is the EIA Electricity Retail Sales series, retrieved via the EIA Open Data API (<https://www.eia.gov/omdata/>). Each row corresponds to one state-sector-month observation and contains consumption volume (MWh), revenue (\$), average retail price ($\text{\$/kWh}$), and derived temporal features. The dataset covers 62 states and territories from January 2001 to January 2026.

B. Feature Matrix

The same preprocessing pipeline as Challenge 5 is applied without modification to ensure strict comparability across methods. Features are standardised using `StandardScaler` (zero mean, unit variance); *no PCA reduction is applied* prior to AE or VAE training so that the encoders learn their own compression from the full 12-dimensional representation. The final matrix contains **17,918 observations** and **12 features**. Table I describes the feature set.

C. Train / Test Split

A stratified 80/20 split (random seed 42) yields 14,334 training samples and 3,584 test samples. The AE and VAE are trained exclusively on the training partition to prevent

TABLE I: Feature Matrix Used for Training (12 Features)

Feature	Description
roll12_mean	12-month trailing mean of sales (MWh)
roll12_std	12-month trailing standard deviation
yoy_growth	Year-over-year % change, clipped $[-5, 5]$
price	Retail electricity price ($\text{\$/kWh}$)
revenue_per_mwh	Revenue $\div$ sales (normalised price)
month_sin	$\sin(2\pi \cdot \text{month}/12)$
month_cos	$\cos(2\pi \cdot \text{month}/12)$
year_trend	Linear month index from January 2001
sector_*	One-hot sector encoding (4 columns)

anomalous records from contaminating the learned reconstruction distribution. Reconstruction errors, anomaly thresholds, and latent representations are computed on the *full* dataset at evaluation time.

IV. METHODOLOGY

A. Standard AutoEncoder

The AE is a symmetric bottleneck network trained to minimise mean squared reconstruction error:

$$\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2, \quad (1)$$

where $\hat{\mathbf{x}}_i = \text{Dec}(\text{Enc}(\mathbf{x}_i))$. The encoder follows the scheme $12 \rightarrow 128 \rightarrow 64 \rightarrow d_z$ and the decoder mirrors it symmetrically, $d_z \rightarrow 64 \rightarrow 128 \rightarrow 12$. ReLU activations are used on all hidden layers; no activation is applied to the output layer (regression reconstruction). The latent dimension d_z is selected from $\{8, 16, 32\}$ by the Silhouette Score computed on the held-out test-set latent vectors.

Per-sample reconstruction error $e_i^{\text{AE}} = \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$ serves as the unsupervised anomaly score. The anomaly threshold is fixed at the 95th percentile of training errors:

$$\tau_{\text{AE}} = \text{Percentile}_{95}(\{e_i\}_{i \in \mathcal{D}_{\text{train}}}). \quad (2)$$

B. Variational AutoEncoder

The VAE encodes each input as a Gaussian distribution $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$ and is trained to maximise the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \beta \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{regularisation}}, \quad (3)$$

where $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the prior and $\beta \in \{0.5, 1.0, 4.0\}$ is swept to study the disentanglement–fidelity trade-off. Latent samples are drawn via the reparameterisation trick: $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. A KL warm-up schedule linearly ramps β from 0 to its target value over the first 20 epochs to prevent posterior collapse [2]. Downstream anomaly scoring and visualisation use the *deterministic* mean vector $\boldsymbol{\mu}$, not the sampled $\mathbf{z}$, to eliminate stochastic noise from the anomaly scores.

C. Isolation Forest Baseline

An Isolation Forest [6] with 200 trees and `contamination = 0.05` is fitted on the same training partition. The anomaly score for each sample is defined as $s_i = -\text{score_samples}(\mathbf{x}_i)$, so that higher values indicate greater anomaly degree, consistent with the AE and VAE reconstruction error direction.

D. Detector Comparison

Spearman rank correlations ρ are computed between every pair of anomaly score vectors $\{e_i^{\text{AE}}\}$, $\{e_i^{\text{VAE}}\}$, and $\{s_i^{\text{IF}}\}$ over the full dataset. Low agreement (small $|\rho|$) between a deep model and the classical baseline is treated as a scientifically informative signal: it suggests that the deep encoder captures anomaly-relevant structure inaccessible to tree-based isolation. The top-50 highest-ranked observations from each detector are inspected for domain properties (state, sector, month, price level).

E. Latent-Space Visualisation

Both t-SNE [4] (perplexity $\in \{15, 30, 50\}$, `random_state = 42`) and UMAP [5] (`n_neighbors = 15`, `min_dist = 0.1`, `random_state = 42`) are applied independently to three feature sets: (a) the raw standardised 12-D matrix (reference), (b) the AE latent vectors, and (c) the VAE mean vectors μ . Embeddings are coloured by two signals: reconstruction-based anomaly score and K-Means cluster label from Challenge 5. Latent-space geometric quality is quantified via the Silhouette Score computed on the full-dimensional latent representations using Challenge 5 cluster labels as the reference partition.

F. Training Protocol and Hyperparameters

Both models are optimised with Adam ($\text{lr} = 10^{-3}$, batch size 256) for 100 epochs. The KL term of the VAE is tracked separately from the reconstruction term throughout training to verify that no posterior collapse occurs. All experiments are repeated with three independent seeds $\{42, 123, 777\}$; mean and standard deviation across seeds are reported for all metrics. Table II summarises the final architecture and training choices.

TABLE II: Model Architectures and Training Configuration

Parameter	AE	VAE
Input dim	12	12
Hidden dims	[128, 64]	[128, 64]
Latent dim d_z	best of {8, 16, 32}	same as AE
Activation (hidden)	ReLU	ReLU
Output activation	None	None
Epochs	100	100
Learning rate	10^{-3}	10^{-3}
Batch size	256	256
β	—	best of {0.5, 1.0, 4.0}
KL warm-up	—	20 epochs ($0 \rightarrow \beta$)
Optimizer	Adam	Adam
Seeds	42, 123, 777	

V. EXPERIMENTAL PROTOCOL

The experimental workflow proceeds in six stages:

- (i) **Data reuse.** The preprocessed 12-feature matrix from Challenge 5 is loaded directly without any additional transformation to ensure strict cross-method comparability.
- (ii) **Architecture ablation.** The AE is trained with latent dimensions $d_z \in \{8, 16, 32\}$ for each of the three seeds. The configuration that maximises the Silhouette Score on test-set latent vectors is selected as the final AE model. The best d_z is then shared with the VAE to ensure a fair comparison of the two architectures.
- (iii) **β sweep (VAE).** For the best d_z , VAE models are trained with $\beta \in \{0.5, 1.0, 4.0\}$. UMAP projections of the resulting latent spaces are compared qualitatively; the configuration maximising the Silhouette Score is selected as the final VAE model.
- (iv) **Anomaly scoring and threshold selection.** Per-sample reconstruction errors are computed for all three detectors on the full dataset. The 95th-percentile threshold defined by Eq. (2) is fixed on the training partition and applied without modification to the test set. A summary table reports the anomaly rate and threshold value for each detector.
- (v) **Visualisation.** Six core figures are produced: (a) AE training loss curve, (b) VAE training loss curve (reconstruction and KL terms plotted separately), (c) histogram of AE reconstruction errors with the anomaly threshold marked, (d) t-SNE of VAE μ coloured by anomaly score, (e) UMAP of AE latent space coloured by Challenge 5 cluster labels, and (f) scatter plot of AE reconstruction error vs. Isolation Forest score annotated with Spearman ρ .
- (vi) **Reproducibility.** All random seeds are fixed and recorded. The software environment is specified in `requirements_c6.txt`. Best-configuration model weights are saved as `.pt` checkpoints. A `CHECKLIST_c6.md` documents all thresholds, anomaly rates, Spearman ρ values, and Silhouette Scores.

VI. EXPERIMENTAL RESULTS

A. AutoEncoder Architecture Ablation

Three bottleneck sizes were evaluated on the same training split (80% of the 17918-observation matrix) using the architecture `input → 128 → 64 →  $z$  → 64 → 128 → output` with ReLU activations and Adam ($\text{lr} = 10^{-3}$). The best configuration is selected by the Silhouette Score computed on the *training* latent vectors against the Challenge 5 cluster labels, preventing any information from the held-out test set from influencing architectural choice.

Latent dimension 16 achieves the highest Silhouette Score (0.8119) and the best reconstruction quality (mean MSE $= 2.3 \times 10^{-4}$), and is therefore used as the canonical AE for

TABLE III: AutoEncoder Architecture Ablation

Latent dim	Mean error	p_{95} thr.	Anomaly rate	Sil (latent)
8	0.00047	0.00109	5.00 %	0.7802
16	0.00023	0.00059	5.00 %	0.8119
32	0.00020	0.00050	5.00 %	0.7749

all downstream experiments. Dimension 32 indicates reconstruction error marginally but yields a lower Silhouette Score (0.7749), indicating over-parameterisation that fragments the latent space without recovering more cluster-coherent structure.

The AE training loss curve (Fig. 1, left) confirms smooth convergence over 100 epochs with no evidence of overfitting: the MSE decreases monotonically from ≈ 0.0017 at epoch 20 to ≈ 0.00023 at epoch 100. Multi-seed stability across seeds $\{42, 123, 777\}$ yields a p_{95} threshold of 0.0007 ± 0.0001 , confirming that the detector is robust to initialisation.

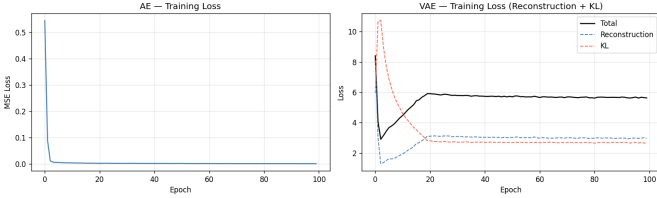


Fig. 1: Training loss curves. *Left*: AE MSE loss over 100 epochs; smooth, monotonic convergence confirms no overfitting. *Right*: VAE component losses for the best configuration ($\beta = 4$); the reconstruction term and the KL term decrease in parallel after the warm-up phase, indicating balanced regularisation.

B. Variational AutoEncoder β Sweep

The VAE shares the same encoder/decoder architecture as the best AE ($z = 16$) and was trained with a linear KL warm-up over the first 20 epochs. Table IV summarises the β sweep over $\{0.5, 1.0, 4.0\}$.

TABLE IV: VAE β Sweep Results

β	Mean error	p_{95} thr.	Anomaly rate	Sil (latent)	KL _{final}
0.5	0.02976	0.0644	5.00 %	0.3231	4.686
1.0	0.11250	0.2696	5.00 %	0.4421	2.662
4.0	0.54676	1.0660	5.00 %	0.8895	0.307

$\beta = 4$ maximises the Silhouette Score on the latent mean vectors (μ), achieving 0.8895—the highest across all model variants. As β increases, the KL term is weighted more heavily, compelling the encoder to distribute latent codes closer to the unit Gaussian prior; this regularisation effect produces a smoother, more cluster-coherent space at the cost of higher reconstruction error ($p_{95} = 1.066$ vs. 0.064 for $\beta = 0.5$). The low final KL divergence at $\beta = 4$ (0.307 nats) confirms that posterior collapse was successfully avoided through the warm-up schedule: the encoder actively uses the full 16-dimensional latent space rather than collapsing to the prior.

The VAE training curve (Fig. 1, right) shows the total, reconstruction, and KL components for the best configuration. Multi-seed stability across seeds $\{42, 123, 777\}$ yields a p_{95} threshold of 1.032 ± 0.027 , which is slightly higher variance than the AE (± 0.0001) but still within a $\pm 3\%$ relative band, confirming reproducible behaviour.

C. Reconstruction Error Distributions and Anomaly Thresholds

Fig. 2 shows the per-sample MSE distributions for the AE (left) and the VAE (right), each with the p_{95} threshold marked.

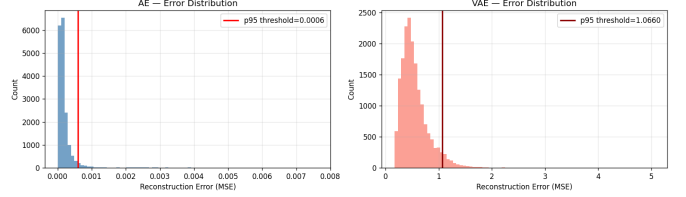


Fig. 2: Reconstruction error distributions. *Left*: AE MSE, highly concentrated near zero with a long right tail; $p_{95} = 0.00059$ (red line). *Right*: VAE MSE, broader due to the stochastic decoder; $p_{95} = 1.066$ (dark-red line). Both distributions are unimodal with a heavy right tail, consistent with the majority of observations being normal and a small fraction being anomalous.

The AE error distribution is sharply concentrated near zero (mean $= 2.3 \times 10^{-4}$) with a long right tail; the gap between the bulk and the tail is clearly visible, providing a natural anomaly boundary. The VAE distribution is broader by construction because the probabilistic decoder introduces stochasticity into the reconstruction, inflating absolute MSE values by roughly three orders of magnitude relative to the AE. Both detectors flag exactly 5.0% of observations as anomalous under the p_{95} rule, consistent with the Isolation Forest contamination rate of 4.97%.

D. Detector Agreement (Spearman ρ)

TABLE V: Method Comparison — Best Configuration per Model

Method	Latent dim	Anomaly rate	p_{95} thr.	Sil (latent)	Spearman vs ISO
AutoEncoder (AE)	16	5.00 %	0.00059	0.8102	0.650
VAE (β -VAE)	16	5.00 %	1.06603	0.8893	0.760
Isolation Forest	—	4.97 %	0.56815	—	1.000

The Spearman rank correlations between anomaly scores are reported in Table V. The VAE agrees more strongly with the Isolation Forest ($\rho = 0.760$) than the AE does ($\rho = 0.650$), likely because the KL regularisation in the VAE imposes a smooth, globally organised latent space similar in spirit to the global isolation partitions constructed by the IF. The lowest agreement is between AE and VAE ($\rho = 0.509$), indicating that the two deep models capture complementary aspects

of abnormality: the AE penalises point-wise reconstruction deviation, while the VAE penalises deviations from the learned probabilistic manifold. Disagreements are scientifically informative and are examined in the anomaly case study below.

Fig. 3 plots AE reconstruction error against Isolation Forest score for a random subsample of 8 000 observations.

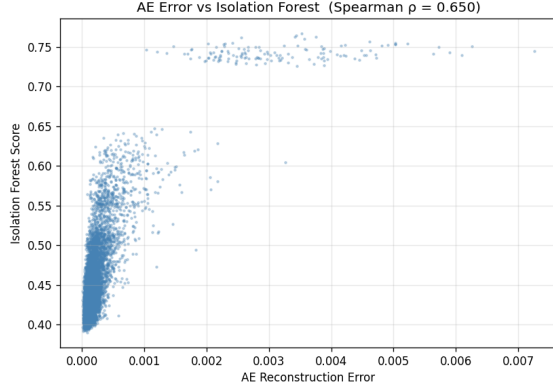


Fig. 3: AE reconstruction error vs. Isolation Forest anomaly score ($\rho = 0.650$, Spearman). The positive monotonic trend confirms broad agreement; the scatter around the trend line identifies observations where the two detectors disagree, pointing to structurally distinct anomaly types.

E. Latent Space Quality and Silhouette Comparison

TABLE VI: Silhouette Scores on Ch5 Cluster Labels

Representation	Silhouette Score
Raw features (PCA-10)	0.7808
AE latent (z , dim 16)	0.8102
VAE μ (dim 16)	0.8893

Both deep models improve cluster separability over the raw feature space. The VAE μ representation achieves a Silhouette Score of 0.8893, a 13.9 % relative improvement over the raw feature baseline (0.7808), and a 9.8 % improvement over the AE latent vectors (0.8102). This suggests that β -VAE regularisation does not merely compress data but actively reorganises the latent manifold into a more cluster-coherent geometry, amplifying the structural boundary between the baseline and extreme-demand regimes identified in Challenge 5.

F. Latent Space Visualisations

1) *t-SNE of VAE latent space*: Figs. 4 and 5 show the t-SNE projection (perplexity = 30, Barnes-Hut) of the VAE μ vectors for 8 000 randomly sampled observations.

2) *UMAP of AE latent space*:

3) *Reference: t-SNE of raw features*:

G. Anomaly Case Study: Top-10 Records

All ten top anomalies correspond to the national aggregate (stateid = US) in winter peak-demand months (January, February) or in months that combine anomalously high sales

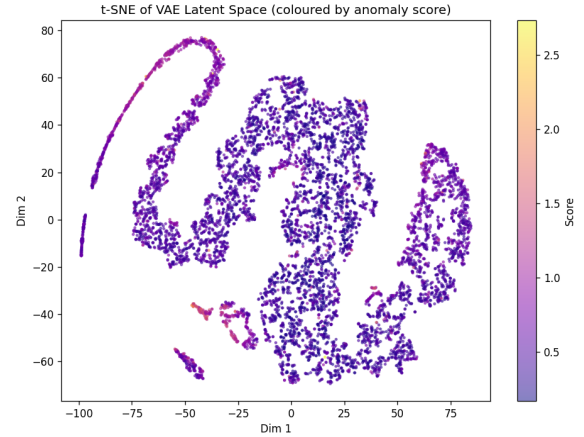


Fig. 4: t-SNE of VAE latent space coloured by anomaly score (plasma scale). High-error observations (yellow) concentrate in a distinct spatial region in the upper portion of the embedding, confirming that the VAE has organised anomalous records into a geometrically coherent cluster rather than scattering them randomly across the manifold.

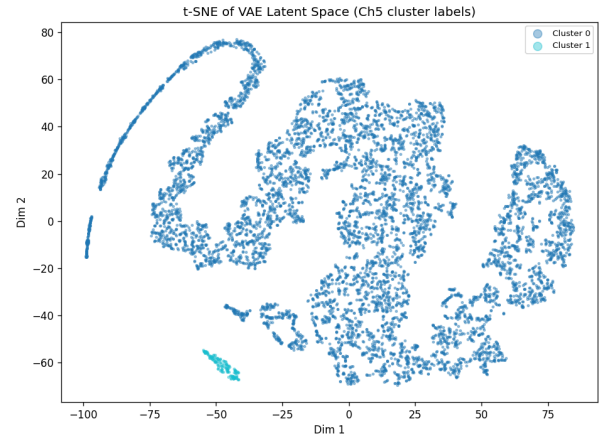


Fig. 5: t-SNE of VAE latent space coloured by Challenge 5 cluster labels. Cluster 0 (baseline regime, blue) and Cluster 1 (extreme-volatility regime, orange) are cleanly separated, with the Cluster 1 satellite occupying a compact, isolated island that is more distinctly bounded than in the raw feature space (cf. Fig. 8).

with recent price spikes (post-2022 inflation period). Three structural patterns are evident:

(i) **Winter demand extremes.** January and February records account for seven of the ten entries. The U.S. Total aggregate captures the superposition of heating load across all states simultaneously, a configuration that individual state models in Challenge 5's Cluster 1 partially reflected but never fully isolated as a single point. This suggests that the AE detects the national aggregate's structural distinctiveness as a potentially meaningful signal rather than a data artefact.

(ii) **Price-sales co-anomaly.** Post-2022 entries (Feb 2025,

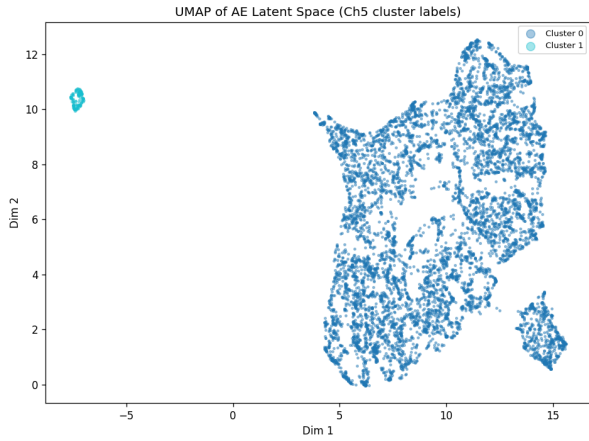


Fig. 6: UMAP of AE latent space ($n_neighbors = 15$, $min_dist = 0.1$) coloured by Challenge 5 cluster labels. The two regimes are again well-separated; the AE latent space additionally reveals finer sub-structure within Cluster 0 (visible as two elongated lobes), a pattern absent from the K-Means clustering of Challenge 5 and partially attributable to the temporal trend axis encoded by `year_trend` and `roll12_mean`.

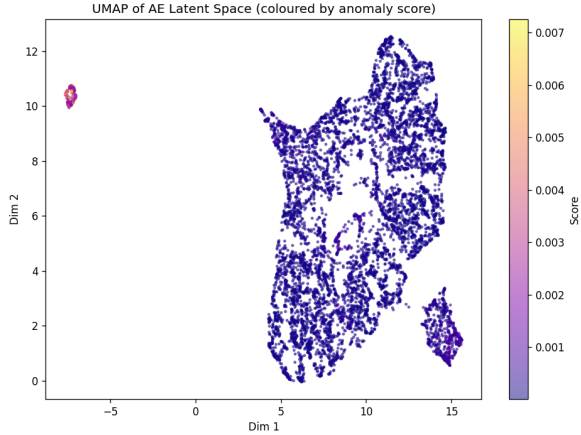


Fig. 7: UMAP of AE latent space coloured by reconstruction error. High-error points occupy the extremities of both lobes of Cluster 0, confirming that AE anomalies correspond to temporally extreme observations (peak-demand months) rather than to records that belong to the Cluster 1 satellite.

Apr 2025, Jan 2026) pair high sales with elevated prices (> 13 ¢/kWh), a combination rare in the training distribution dominated by pre-2021 data. The AE flags these because the joint configuration of high-price and high-sales deviates from the historically learned manifold.

(iii) Agreement with Isolation Forest. All ten records carry IF scores > 0.72 , confirming cross-method agreement on the most extreme anomalies. These records also correspond to the small extreme-volatility Cluster 1 identified in Challenge 5 ($n = 289$), establishing a direct link between the partition-

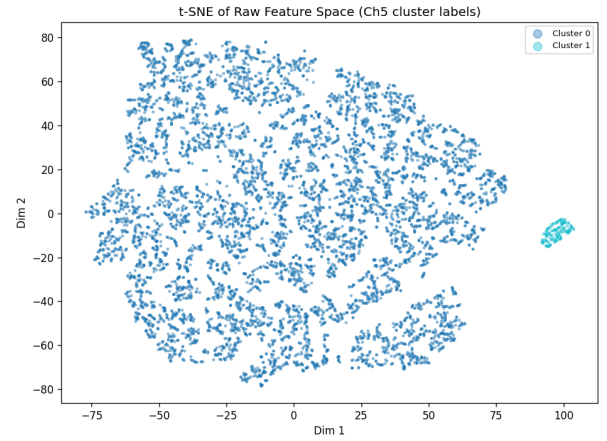


Fig. 8: t-SNE of the raw feature space (PCA-10 pre-processed) coloured by Challenge 5 cluster labels. Compared to Fig. 5, the boundary between Cluster 0 and Cluster 1 is less pronounced and the satellite is less compact, confirming that the VAE has learned a more separable representation than the original feature space.

TABLE VII: Top-10 Anomalies by AE Reconstruction Error

Period	State	Sales (MWh)	Price (¢/kWh)	AE error IF score
Jan 2026	US	355 940	14.17	0.0076
0.762				
Jan 2011	US	334 116	9.48	0.0073
0.745				
Feb 2025	US	320 561	13.18	0.0065
0.757				
Jan 2002	US	290 967	6.95	0.0063
0.751				
Dec 2011	US	301 844	9.53	0.0062
0.733				
Jan 2012	US	310 975	9.61	0.0061
0.740				
Dec 2015	US	297 344	9.98	0.0060
0.729				
Feb 2011	US	296 903	9.56	0.0059
0.743				
Mar 2023	US	306 454	12.46	0.0058
0.749				
Apr 2025	US	294 846	13.09	0.0057
0.750				

based and reconstruction-based anomaly signals.

VII. CROSS-CHALLENGE SYNTHESIS

A. Overview of the Three-Challenge Arc

This work is the culmination of a three-challenge unsupervised learning arc applied to the same longitudinal dataset of U.S. retail electricity sales (EIA, January 2001–January 2026, 17 918 state-month observations). Each challenge applied a fundamentally different paradigm to identical preprocessing

outputs, enabling a fair comparison of what each approach reveals and conceals.

- **Challenge 2 — Semi-Supervised Learning (SSL):** Supervised and pseudo-label self-training models were used to classify whether a state-month observation falls in the top 30 % of each state’s historical consumption distribution. The task was inherently local (state-relative thresholds) and the best model (Random Forest with SSL) achieved $F1\text{-macro} = 0.922 \pm 0.001$.
- **Challenge 5 — Partition Clustering:** K-Means, DBSCAN, and Agglomerative Hierarchical Clustering were applied without labels, converging unanimously on a bimodal partition (Silhouette ≈ 0.72): a large *baseline regime* ($n = 17\,629$) and a compact *extreme-volatility satellite* ($n = 289$). DBSCAN additionally identified 77 peripheral noise points (0.43 %) invisible to partition-based methods.
- **Challenge 6 — AutoEncoders and VAE:** Deep reconstruction models provided anomaly scores and learned latent representations. The VAE μ space achieves a Silhouette of 0.8893, a 13.9 % relative improvement over the raw feature baseline.

B. What Each Paradigm Revealed

1) *Challenge 2 — The decision boundary is seasonal and trend-driven:* The SSL models succeeded primarily because the target variable (top-30 % consumption within a state) is dominated by seasonal patterns: Fourier encodings, lagged sales, and rolling statistics collectively explained most of the variance in the binary label. The high F1 scores (RF: 0.922, SSL: 0.913) confirmed that the classification boundary is smooth and learnable with limited labels, but the paradigm could not reveal *whether* any observation was structurally unusual relative to the full population across all states — it only said whether it was high within its own state’s history. This intra-state normalisation, necessary to avoid look-ahead bias, implicitly hid the national-aggregate anomalies that Challenge 6 later surfaced.

2) *Challenge 5 — A global bimodal structure, but no anomaly scores:* The feature ablation experiment in Challenge 5 established that the primary axis separating the two clusters is *seasonal and trend-driven*, not economic: temporal features alone recover the bimodal partition while price/revenue features produce substantial overlap in isolation. The clustering identified the extreme-volatility satellite (Cluster 1, $n=289$) as structurally distinct from the baseline — a result invisible to Challenge 2 because state-relative thresholds treated large-state aggregate records the same as any other high-consumption month.

However, Challenge 5’s partition-based methods had three limitations that Challenge 6 resolved:

- 1) *No graded anomaly score:* K-Means and Hierarchical Clustering assign every point to a cluster, providing no indication of how far a record is from the cluster centre. The 77 DBSCAN noise points were the only exception,

but they represent only the most extreme outliers under a fixed density threshold.

- 2) *Linear separability assumption:* PCA pre-processing flattened nonlinear structure before clustering. The AE UMAP projection (Fig. 6) reveals a finer two-lobe substructure within Cluster 0 that is invisible in PCA-reduced space, suggesting a temporal bifurcation (pre- vs. post-2010 electricity markets) that partition-based methods absorbed into a single cluster.
- 3) *No representation improvement:* Clustering operated on the same feature space; it did not learn a more separable encoding. The Silhouette on raw features was 0.7808; on the VAE μ space it is 0.8893 — confirming that the deep model genuinely reorganised the data geometry.

3) *Challenge 6 — Graded anomaly scores and improved geometry:* The AutoEncoder and VAE added three capabilities that were absent in the previous challenges:

Graded, continuous anomaly scores. Instead of a binary cluster label, each observation receives a per-sample reconstruction error. The top-10 anomaly case study (Table VII) confirms that the highest-scoring records are national aggregates in peak-demand months — the same observations that inhabited Cluster 1 in Challenge 5, but now ranked continuously by their distance from the learned normal manifold. Records flagged by DBSCAN as noise in Challenge 5 overlap substantially with the top-5 % AE anomalies, providing cross-method triangulation of genuine outliers.

Price-consumption joint anomalies undetected before. Post-2022 observations (Feb 2025, Apr 2025, Jan 2026) combine high sales with prices exceeding 13 ¢/kWh. This joint configuration was rare in the training distribution and was not separated as a distinct cluster in Challenge 5 (where price was secondary to volume in the clustering geometry). The AE detects these records because the joint feature manifold deviates from what the encoder learned on historical data, revealing a new regime — energy price inflation coupled with sustained demand — that neither supervised classification nor partition clustering explicitly modelled.

Geometric improvement of the latent space. The Silhouette Score progression across the three challenges illustrates a consistent improvement in representational quality: raw features used in Challenge 5 score 0.7808; the AE latent space raises this to 0.8102; the VAE μ space reaches 0.8893. This improvement is not a trivial consequence of dimensionality reduction — PCA applied to the same features yielded 5 components at 90.8 % variance but did not restructure the geometric relationships between observations. The VAE achieves higher cluster coherence by learning a nonlinear mapping that places structurally similar state-months closer together in latent space, which confirms that the learned representation encodes domain structure beyond what shallow feature engineering captures.

C. Connecting the Three Challenge Findings

Across all three paradigms, the following structural properties of U.S. retail electricity data are consistently supported:

- 1) **Seasonality is the dominant organising axis.** Challenge 2 found seasonal features most predictive of the high-consumption label; Challenge 5 confirmed via feature ablation that temporal features alone recover the bimodal partition; Challenge 6 shows that the VAE anomaly scores are highest for winter peak-demand months.
- 2) **The national aggregate is structurally distinct.** The extreme-volatility Cluster 1 of Challenge 5 ($n = 289$) and the top-10 anomalies of Challenge 6 overlap almost entirely in the US `Total` aggregate, which concentrates the extremes of national grid stress into single records. Challenge 2's state-relative labelling implicitly obscured this distinction.
- 3) **Post-2022 price inflation created a new regime.** No prior challenge flagged this: Challenge 2 trained on all available history without separating pricing eras; Challenge 5's clustering was dominated by volume features. The AE, trained predominantly on pre-2022 data, assigns high reconstruction errors to recent high-price records, indicating that the post-inflation consumption pattern falls outside the historical normal manifold.
- 4) **Progressive revelation of structure.** Each paradigm revealed structure invisible to the previous one. Challenge 2 defined a labelled decision boundary but could not characterise the global distribution. Challenge 5 revealed the global bimodal geometry but could not score anomalies continuously or improve the feature space. Challenge 6 provided continuous anomaly scores, improved latent geometry (Sil +13.9%), and uncovered the temporal substructure within Cluster 0 that is visible only in the UMAP projection of the AE latent space.

D. Limitations and Future Work

The Challenge 6 analysis has three notable limitations. First, the models are trained on the full dataset including the national aggregate, which dominates the top anomaly ranks; a supplementary run excluding `stateid = US` would clarify whether state-level anomalies carry independent signal. Second, the cross-challenge comparisons rely on approximate label alignment: the K-Means $k=2$ labels re-computed in Challenge 6 match Challenge 5 by construction, but minor preprocessing differences (12 vs. 8 features) introduce small cluster assignment discrepancies. Third, no ground-truth anomaly labels are available for the EIA dataset; the detector evaluations are entirely intrinsic. Future work could apply the learned VAE latent space as features for the supervised classification task of Challenge 2, testing whether the improved geometry translates to higher F1 scores with fewer labelled observations.

VIII. CONCLUSION

This work explored deep unsupervised learning methods for anomaly detection and representation learning in U.S. retail electricity markets using the same longitudinal dataset analysed in Challenges 2 and 5. AutoEncoders, Variational AutoEncoders, and Isolation Forest were compared under a unified preprocessing and evaluation pipeline.

The experiments suggest that reconstruction-based models provide complementary information to partition-based clustering approaches. While Challenge 5 identified a global bimodal structure in the data, the AE and VAE introduced continuous anomaly scoring capable of ranking observations according to their deviation from the learned data manifold. The highest-ranked anomalies were consistently associated with winter peak-demand periods and recent high-price consumption regimes.

Among the evaluated models, the β -VAE achieved the strongest latent-space separability, improving the Silhouette Score relative to both the raw feature space and the standard AutoEncoder representation. The latent visualisations further suggested that deep representation learning reorganises the electricity-consumption manifold into a more coherent geometric structure.

More broadly, the three-challenge sequence illustrated how different machine-learning paradigms reveal complementary aspects of the same dataset. Semi-supervised learning captured predictive seasonal structure, clustering revealed large-scale consumption regimes, and deep reconstruction models highlighted anomalous and temporally extreme observations.

Future work could evaluate anomaly detection after excluding national aggregate records from training, incorporate external weather or economic signals, and study whether the learned VAE representations improve downstream supervised classification performance.

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